WORKED SOLUTION

STEP I 2008, Question 2

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q2

The key idea

The substitution is chosen so that the square root becomes $t - x$, which is simple. Rather than solving for x and differentiating a messy quotient, we square to remove the root and then differentiate the resulting relation implicitly. The derivative contains exactly the factor $t - x$ that cancels the square root in the integrand, so the integral collapses to $\int \frac{1}{t+b}\,\mathrm{d}t$. The final part is a test of when that cancellation and division are legitimate.

Solution

The derivative

Write $R = \sqrt{x^2 + 2bx + c}$, so that $t - x = R$. Squaring,

$$t^2 - 2tx + x^2 = x^2 + 2bx + c \quad \Longrightarrow \quad t^2 - 2tx - 2bx - c = 0.$$

Differentiating this with respect to t (with x a function of t),

$$2t - 2x - 2t\frac{\mathrm{d}x}{\mathrm{d}t} - 2b\frac{\mathrm{d}x}{\mathrm{d}t} = 0, \qquad \text{so} \qquad (t+b)\frac{\mathrm{d}x}{\mathrm{d}t} = t - x.$$

To divide by $t + b$ we need it to be non-zero. Since $b^2 < c$,

$$x^2 + 2bx + c = (x+b)^2 + (c - b^2) > (x+b)^2,$$

so $R > |x + b|$ and therefore

$$t + b = (x + b) + R > (x + b) + |x + b| \geq 0.$$

Hence $t + b > 0$ for every x, and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{t - x}{t + b},$$

as required. (The squared relation can also be solved directly: $x = \dfrac{t^2 - c}{2(t+b)}$, and differentiating this gives the same answer after some algebra.)

The integral

The quadratic $x^2 + 2bx + c$ is positive for all x, so the integrand is defined everywhere. Substituting, and using $R = t - x$,

$$\int \frac{1}{\sqrt{x^2+2bx+c}}\,\mathrm{d}x = \int \frac{1}{t-x}\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int \frac{1}{t-x}\cdot\frac{t-x}{t+b}\,\mathrm{d}t$$
$$= \int \frac{1}{t+b}\,\mathrm{d}t = \ln(t+b) + C,$$

where no modulus is needed because $t + b > 0$. The cancellation of $t - x = R$ is fine because $R > 0$. Returning to x,

$$\int \frac{1}{\sqrt{x^2+2bx+c}}\,\mathrm{d}x = \ln\left(x + b + \sqrt{x^2+2bx+c}\right) + C.$$

You can confirm this by differentiating: the derivative of $x + b + R$ is $1 + \dfrac{x+b}{R} = \dfrac{R+x+b}{R}$, and dividing by $x + b + R$ leaves $\dfrac{1}{R}$.

The case $b^2 = c$

Now $x^2 + 2bx + c = (x+b)^2$, so $\sqrt{x^2+2bx+c} = |x+b|$ and the integrand is $\dfrac{1}{|x+b|}$ (defined for $x \neq -b$).

Ask yourself Why is $\sqrt{(x+b)^2}$ not simply $x + b$?

The square root symbol means the non-negative root, so $\sqrt{(x+b)^2} = |x+b|$, which equals $-(x+b)$ when $x + b < 0$. This sign is exactly what decides whether the formula survives.

If $x + b > 0$: directly,

$$\int \frac{1}{x+b}\,\mathrm{d}x = \ln(x+b) + C.$$

The formula gives

$$\ln\left(x + b + (x+b)\right) = \ln 2 + \ln(x+b),$$

which differs from the direct answer only by the constant $\ln 2$. So the result still holds.

If $x + b < 0$: directly,

$$\int \frac{1}{-(x+b)}\,\mathrm{d}x = -\ln\left(-(x+b)\right) + C = -\ln|x+b| + C.$$

But the formula gives $\ln\left(x + b + |x+b|\right) = \ln\left((x+b) - (x+b)\right) = \ln 0$, which is undefined. So the result does not hold here.

It is worth seeing where the derivation breaks. With $x + b < 0$ we have $t = x - (x+b) = -b$ for every x: t is constant, so it cannot be used as a new variable at all, and $t + b = 0$

is exactly the quantity we divided by. The strict inequality $b^2 < c$ was what guaranteed $t + b > 0$. When $b^2 = c$ only the weak inequality $t + b \geq 0$ survives, with equality precisely when $x + b \leq 0$.

Checks

- With $b = 0$, $c = 1$ the result reads $\int \frac{1}{\sqrt{x^2+1}}\,\mathrm{d}x = \ln\left(x + \sqrt{x^2+1}\right) + C = \operatorname{arsinh} x + C$, the standard result.
- Numerically, with random b and $c > b^2$, Simpson's rule for the integral between random limits agrees with the difference of the log formula to better than 10^{-7}. For $b^2 = c$ with $b = 1.3$, $\int_{0.5}^{4} \frac{\mathrm{d}x}{|x+b|} \approx 1.07992$, matching $\ln \frac{5.3}{1.8}$ from the formula.

Common slips

- Writing $\ln|t + b|$ without noticing that $t + b > 0$, and so missing what goes wrong when $b^2 = c$.
- Taking $\sqrt{(x+b)^2} = x + b$ in the final part, which hides the failure for $x + b < 0$ entirely.

Beyond STEP

The substitution $t = x + \sqrt{x^2 + 2bx + c}$ is one of *Euler's substitutions*, which turn any integral of a rational function of x and $\sqrt{ax^2 + bx + c}$ into the integral of a rational function of t. The final part illustrates why a change of variable needs the new variable to depend smoothly and one-to-one on the old one: when t is constant the method collapses. Bartle and Sherbert, *Introduction to Real Analysis*, states the change-of-variables theorem with these hypotheses made precise.

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